

Lab 06 – Detection and Segmentation reflections.

Concepts:

1. *Explain the difference between classification, detection, and segmentation in your own words. Give one workforce example of each.*
 - In computer vision, classification tells what the whole image is. In context, for example, a factory quality system might classify one product photo as either good or defective. Detection tells us what objects are in the image and where these objects are located by drawing boxes around them. An example of this is where a parking lot system can detect and count cars in different parking spaces. Segmentation goes further by identifying the exact pixels that belong to each object. For example, medical imaging systems can outline the exact shape of an organ instead of just drawing a box around it.
2. *What is Non-Maximum Suppression, and why does YOLO use it?*
 - Non-Maximum Suppression, or NMS, is a cleanup step after detection. NMS models may create several overlapping boxes around the same object, while keeping the box with the highest confidence score/rate and removing the extra overlapping boxes. YOLO uses NMS to provide a cleaner, final result and doesn't show many duplicate objects for one object.
3. *In your own words, what does IoU measure?*
 - IoU is an abbreviation for Intersection over Union. The IoU measures how much the predicted box overlaps with the correct ground-truth box. A high IoU score indicates that the model had located an object with high accuracy, while low IoU scores tell us that the predicted box didn't line up well with the real object.

Hands-on Experience

4. *When you ran YOLO11 on your own image, did it detect everything you expected?*
 - I used the Ultralytics zidane.jpg image for the lab. YOLO11 detected the main people in the image, which I expected, while also detecting smaller details such as a tie or sports-related object depending on confidence. Overall, it

succeeded because the objects are common categories from the COCO dataset, such as “person.” If anything was missed, it was probably because the object was small, partly hidden, or not one of the categories YOLO was trained to recognize.

5. *Detection vs. segmentation: when is a box enough, and when is a mask needed?*

- A box is enough when I only need to know the approximate location of an object such as the number of cars in a parking lot or whether someone is inside a restricted area. Masks are needed if the exact shape matters. For example, if a robot arm needs to pick up an object, or if a photo editing app needs to remove the background around a person, segmentation is better because it gives the exact object outline.

6. *SAM 2 does not output category labels. Why is that important?*

- SAM 2 doesn’t output category labels because SAM 2 can segment almost anything I point to, but it does not tell me what the object is. While YOLO can create labels such as “person,” “bus,” or “car,” SAM 2 only gives the mask. That means SAM 2 is flexible for unusual objects, but I may need another model like YOLO to identify the object name.

Trade-offs

7. *(Confidence threshold trade-off) Give one application where you would want a low threshold (more sensitive) and one application where you would want a high threshold. Explain why.*

- I would use a low confidence threshold for a workplace safety alert system because missing a person in a dangerous area can create serious risks., while I would use a high confidence threshold for an automatic photo-tagging system because wrong tags are most of the time embarrassing and annoying. In this context, it is better to only tag something when the model is very confident.

8. *(Precision versus recall) For a workplace safety system that detects unsafe behavior on a factory floor, do you want higher precision or higher recall? What about for a system that tags products in a customer’s photo upload?*

- For a workplace safety system, I would prefer higher recall because it allows the system to catch as many unsafe situations as possible, even if it creates some false alarms, while for systems that tags products in a customer's photo upload, I would prefer higher precision. It is better to avoid wrong tags because incorrect product labels could confuse customers.
9. *When would you reach for a specialist model like YOLO11-seg, and when would you reach for a foundation model like SAM 2?*
- I would use YOLO11-seg when the objects are common categories that YOLO already knows, such as people, cars, buses, dogs, or chairs, in addition to its high speed and the fact it gives both labels and masks. I would use SAM 2 for tasks required to segment unusual objects or objects outside YOLO's fixed categories. SAM 2 is better when I can give it a prompt, click, or box and ask it to segment whatever I point to.

Midterm project

10. *What project area did you pick for your midterm? In one paragraph, describe: (a) what the application does, (b) where the data will come from, and (c) how you might extend it for the capstone.*
- For my midterm project, I would choose a PPE compliance check system. The application would detect whether workers in factory or construction images are wearing required safety equipment such as helmets, safety vests, and goggles to ensure safety and security at the highest level. The provided data could come from a small public PPE image dataset or from sample workplace safety images that have been collected online for class use. For the capstone version, I could extend the project into a real-time video feed system that tracks workers over time and sends alerts when someone enters a work area without the proper PPE.